Neural Excitation in the FitzHugh-Nagumo Model

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Abstract

The FitzHugh-Nagumo model of neural excitation has stimulated a great deal of interdisciplinary research since its synthesis by biophysicist Richard FitzHugh in 1961. The historical context for its development is briefly summarized here. The equations are studied in some detail, through mathematical analysis and computer simulations. Two extensions of the model are considered: noisy inputs and a small network comprised of two electrically-coupled neurons.

Introduction

Historical Context

The phenomenon of excitability in electrically-active biological systems is a subject of ongoing research. Although the scholarly study of physiology has a long history dating back to the ancient Greeks, Indians, and Egyptians, its prosperous marriage with quantitative methods is a more recent phenomenon.

The physiological process undertaken by neurons and cardiac cells known as an "action potential" is the main focus of this paper. Action potentials are quick (a few milliseconds long) spikes in the voltage of a cell membrane that propagate distances spanning the length of the hosting organism. A student of the famous physicist Hermann von Helmholtz (1821–1894), Julius Bernstein (1839-1917) was a German scientist who is accredited as the first one to provide a plausible description of action potentials. In order to do so, he developed an experimental device called a "differential rheotome" which allowed him to measure the electrical activity in nerves and muscle cells as a function of time. He leveraged his new invention to determine the duration of the various phases of the action potential, as well as an average propagation speed of 28.7 m/s. Bernstein reconciled his observations with Walter Nernst's equation for the equilibrium potential of an ionic solution divided by a barrier, including the predicted temperature dependence of the resting potential. While the Nernst equation explains the **static** properties of the nerve membrane, Bernstein, with his differential rheotome, shed light on the origin of its **dynamics**. This work asserted that the membrane of an excitable cell acts like a selectively-permeable barrier with electrochemical properties that **change** during the course of an action potential. Crucially, Bernstein focused solely on the potassium ions and neglected the flow of sodium ions. This left unknown the "overshoot" of the action potential, during which the neuron is unexcitable [1]. This is known as the "refractory" period. Today, Bernstein has an interdisciplinary research hub rooted in Germany named after him, located virtually at this URL.

The mystery of the refractory period, along with a quantitative basis for many other properties of action potentials, would not be solved until the early 1950's with the seminal work of Alan Hodgkin (1914-1998) and Andrew Huxley (1917-2012). Enabled by the invention of the "voltage clamp" by George Marmont [2] in 1947, and later refinement by Kenneth Cole (1900-1984), Hodgkin and Huxley performed experiments on the electrophysiology of the squid's giant axon. This new experimental technique allowed them to safely neglect axial currents in the neuronal membrane and measure the transverse currents at any fixed voltage [3]. These experiments led Hodgkin and Huxley to develop a framework of neuronal excitation that includes not only the potassium current realized earlier by Bernstein, but also the integral sodium current that initiates the depolarization of the action potential. Accounting for both the sodium and potassium membrane currents, Hodgkin and Huxley came up with a mathematical model for action potential generation in a patch of membrane that consists of four ordinary differential equations (ODE's). Their work [4] has influenced an enormous amount of research; the paper referenced here has been cited 25517 times (as of November 29, 2021), according to Google scholar.

Although remarkable, the Hodgkin-Huxley model is not amenable to the same mathematical analyses as lower dimensional dynamical systems. This observation is what led Richard FitzHugh (1922-2007), a biophysicist working at the National Institutes of Health (NIH) during the 1960's, to develop a 2-dimensional analog to the 4-dimensional Hodgkin-Huxley model of action potential generation [5]. A year after the American scientist FitzHugh published his mathematical model in 1961, a Japanese engineer Jin-Ichi

Nagumo (1926-1999) published work on an electronic circuit he and his colleagues designed that displays dynamics that are similar to those displayed by FitzHugh's equations [6]. This set of ODE's is now referred to as the FitzHugh-Nagumo (FHN) model.

Modern Work

The FitzHugh-Nagumo model is being used for research in excitability and neuronal dynamics, to this day. Researchers have extended it to study spiral waves that occur in the voltage dynamics of the heart preceding fibrillation [7], chimera states in networks of neurons [8], and epilepsy [9], among many others. The extension of the FHN model considered here is that of an electrically-coupled pair of neurons, inspired by the recent work of ref. [10] (see Methods).

Methods

The FitzHugh-Nagumo neuron model is stated as follows:

$$\dot{v} = c\left(v - \frac{v^3}{3} + w + I_{\text{ext}}\right) \tag{1a}$$

$$\tau\dot{w} = -\frac{1}{c}(v - a + bw) \tag{1b}$$

where v represents voltage, w is the activation variable, I_{ext} is the external stimulus current, and $a, b, c > 0$ are constant parameters. The nullclines of the model are the curves on which $\dot{v} = 0$ and $\dot{w} = 0$. Hence, they are given by

$$w = \frac{v^3}{3} - v - I_{\text{ext}} \tag{2a}$$

$$w = \frac{1}{b}(a - v) \tag{2b}$$

Equation 2a above is the curve on which $\dot{v} = 0$ and equation 2b is where $\dot{w} = 0$. Note that the position and shape of the nullclines do not depend on the parameter c. These curves are plotted in figure 1. There is an intersection of the curves for the parameter values used, in this case, which represents a fixed point of the system where both $\dot{v} = 0$ and $\dot{w} = 0$. Solutions to equation 1, referred to here as "trajectories," starting on a fixed point will remain at the fixed point for all time.

In order to examine the stability of an arbitrary fixed point (v_∞, w_∞) analytically, we must first assemble the stability matrix S of partial derivatives [11]:

$$S(v_\infty, w_\infty) = \left.\begin{bmatrix} \frac{\partial \dot{v}}{\partial v} & \frac{\partial \dot{v}}{\partial w} \\ \frac{\partial \dot{w}}{\partial v} & \frac{\partial \dot{w}}{\partial w} \end{bmatrix}\right|_{(v,\,w)\,=\,(v_\infty,\,w_\infty)}$$

$$= \begin{bmatrix} c(1 - v_\infty^2) & c \\ -\frac{1}{\tau c} & -\frac{b}{\tau c} \end{bmatrix}$$

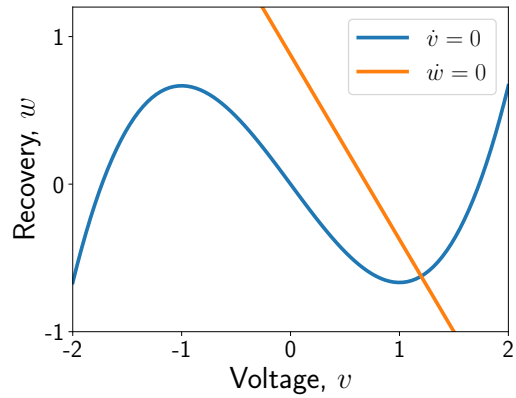

Figure 1: Plot of the nullclines (equation 2), for parameter values $a = 0.7$, $b = 0.8$, $c = 3$, $I_{\text{ext}} = 0$. The intersection of the nullclines corresponds to a fixed point, referred to as (v_∞, w_∞) in the text.

The trace $\text{tr}(S)$ and determinant $\Delta(S)$ of the matrix S, when evaluated at the fixed point (v_∞, w_∞), are indicative of the fixed point's stability. These quantities are given by

$$\text{tr}(S) = c(1 - v_\infty^2) - \frac{b}{\tau c} \tag{3a}$$

$$\Delta(S) = \frac{b}{\tau}(v_\infty^2 - 1) + \frac{1}{\tau} \tag{3b}$$

Consider the case $I_{\text{ext}} = 0$, for which the nullclines are plotted in figure 1. It can be seen graphically that the intersection of the nullclines corresponding to a fixed point has $v_\infty > 1$. Since all of the parameter values are positive, this implies that $\text{tr}(S) < 0$ and $\Delta(S) > 0$ at the fixed point (v_∞, w_∞) corresponding to zero external current $I_{\text{ext}} = 0$. Thus, both eigenvalues of the system linearized about the fixed point are negative, meaning that the fixed point is linearly stable. This is confirmed via numerical simulation of equations 1 using the Dormand-Prince method. Figure 2 shows numerical solutions superposed with the vector field and nullclines of the FitzHugh-Nagumo model. It is clear that small perturbations from the resting state, represented by the intersection of the nullclines in the bottom right of figure 2, shortly return back to the equilibrium. This numerical demonstration confirms the linear stability analysis performed above. In fact, as we increase I_{ext} from 0, the cubic nullcline is shifted down with respect to its position in figures 1 and 2, always resulting in an equilibrium voltage greater than one ($v_\infty > 1$). Thus, for any set of positive parameter values there exists a stable equilibrium, or "resting" state. The phase portrait will therefore look qualitatively like figure 2 for any $I_{\text{ext}} \geq 0$.

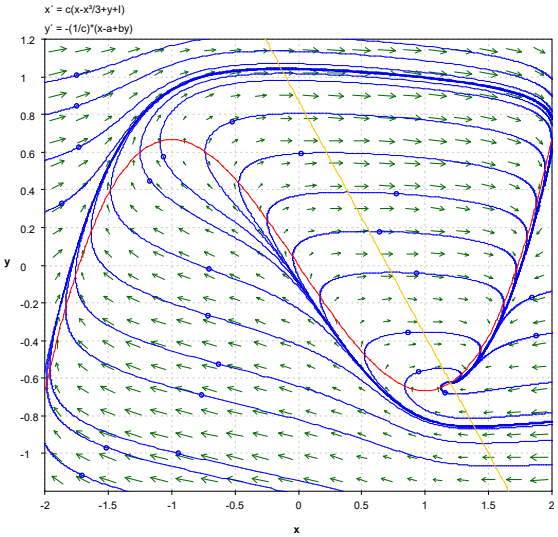

Figure 2: Phase portrait of equations 1 with voltage v on the horizontal axis and recovery variable w on the vertical axis. The v nullcline is shown in red, the w-nullcline in yellow, sample trajectories in blue, and the vector field in green. The length of the green arrows scales with the size of the derivatives at that point. Parameter values are $a = 0.7$, $b = 0.8$, $c = 3$, $I_{\text{ext}} = 0$, $\tau = 1$. The equilibrium is stable here.

What if the stimulus current is negative $I_{\text{ext}} < 0$? In this case, the trace of the stability matrix $\text{tr}(S)$ and determinant $\Delta(S)$ given by equations 3 have to potential to switch sign at the equilibrium. Because of this, we may hypothesize that the equilibrium may change stability when we make $I_{\text{ext}} < 0$. But if trajectories become unstable near the equilibrium, where do they fly off to? They must be bounded. This is a model of a neuron, after all, and it would be a rather poor model if voltage became infinite when the stimulus current was negative.

Indeed, if the stimulus current is negative enough, the resulting trajectories that start anywhere, except for ON the equilibrium, will converge to a limit cycle. A limit cycle is a special kind of periodic trajectory that all non-equilibrium initial conditions end up on after enough time. This is demonstrated numerically via figures 3 and 4, where $I_{\text{ext}} < 0$ is decreased gradually to result in the loss of stability of the resting state and an appearance of a stable limit cycle. This stable limit cycle, visualized as the dark blue loop in figure 4, corresponds to so-called "tonic" spiking in the neuron. All this means is that the neuron spikes with a constant period.

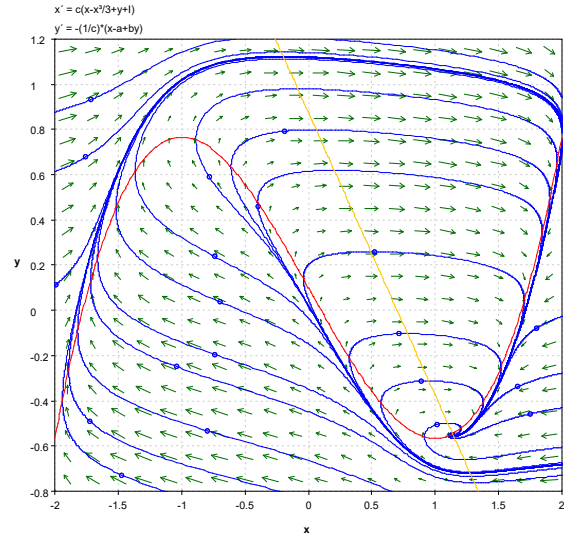

Figure 3: Phase portrait of equations 1 with voltage v on the horizontal axis and recovery w on the vertical axis. Colors and parameter values are as in figure 2, except $I_{\text{ext}} = -0.1$. The equilibrium is stable here.

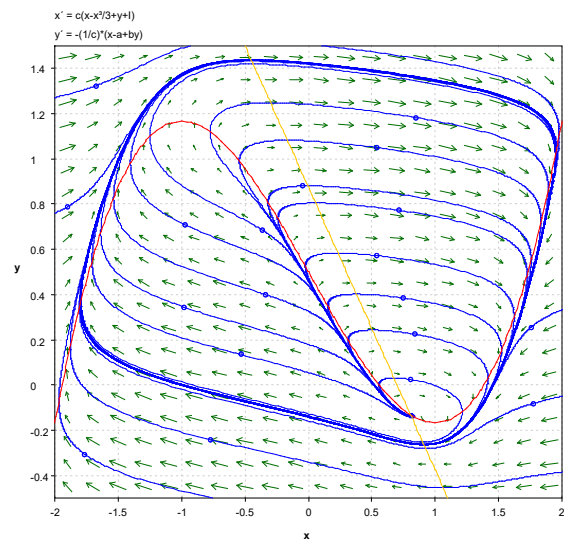

Figure 4: Phase portrait of equations 1 with voltage v on the horizontal axis and recovery w on the vertical axis. Colors and parameter values are as in figure 2, except $I_{\text{ext}} = -0.5$. The equilibrium has lost the stability it had in figures 2 and 3 in a supercritical Hopf bifurcation (see figure 7), resulting in a stable limit cycle corresponding to tonic spiking in the neuron.

However, if I_{ext} is TOO negative, then the equilibrium from earlier regains stability, but at a lower voltage. This is shown through the sequence of figures 5 and 6.

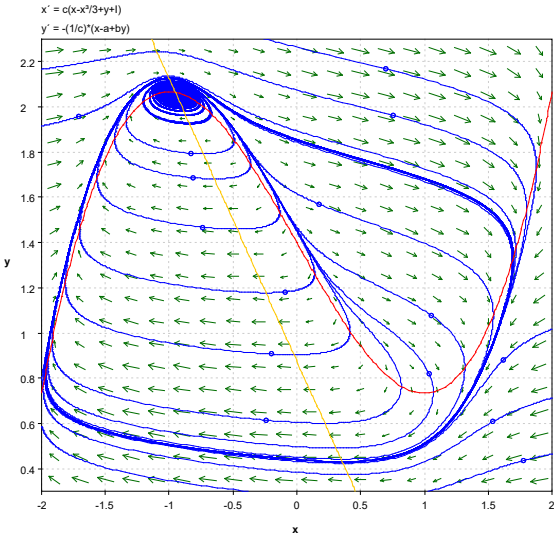

Figure 5: Phase portrait of equations 1. Axes, colors and parameter values are as in figure 2, except $I_{\text{ext}} = -1.4$. The equilibrium is still unstable here, but trajectories take many rotations to escape its neighborhood (dense spirals in the top left) to eventually converge to the limit cycle illustrated by the thick, dark blue loop.

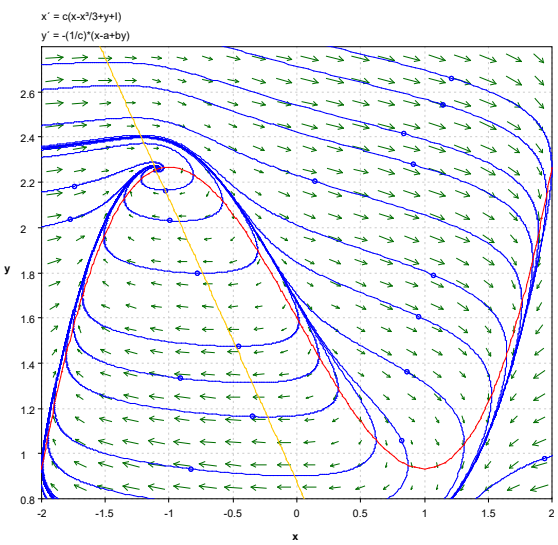

Figure 6: Phase portrait of equations 1. Axes, colors and parameter values are as in figure 2, except $I_{\text{ext}} = -1.6$. The equilibrium has regained stability at the cost of the annihilation of the limit cycle, again through a supercritical Hopf bifurcation (see figure 7).

Two electrically-coupled FHN neurons

The model of equation 1 is extended to the case of two electrically-coupled neurons, with the following governing equations [10]:

$$\dot{v_1} = c\left(v_1 - \frac{v_1^3}{3} + w_1\right) + \gamma_{12}(v_1 - v_2) \tag{4a}$$

$$\tau\dot{w_1} = -\frac{1}{c}(v_1 - a + bw_1) \tag{4b}$$

$$\dot{v_2} = c\left(v_2 - \frac{v_2^3}{3} + w_2\right) + \gamma_{21}(v_2 - v_1) \tag{4c}$$

$$\tau\dot{w_2} = -\frac{1}{c}(v_2 - a + bw_2) \tag{4d}$$

The subscript 1 is for neuron 1 and the subscript 2 is for neuron 2. The externally applied current from equation 1 has been replaced by a coupling to the opposing neuron, where γ_{12} and γ_{21} are the coupling strengths. This model is studied numerically in the results section below.

Results

Excitation Block

The phenomenon of "excitation block" [12] is demonstrated in figures 7 and 8 below with the single-neuron model. Excitation block is a process whereby the input stimulus is raised so high that the neuron does not spike anymore, but rather tends to an elevated constant voltage. In the bifurcation diagram of figure 7, this is shown by plotting the asymptotic voltage values for FHN neurons, given random initial conditions, against the injected DC current. For currents $I_{ext} < -1.4$ or so, the neuron always tends to a low, stable voltage, as indicated by the black dots on the left of the figure. This regime corresponds to the trajectories shown in figure 6, where the neuron is excitable but no stable limit cycle exists. For currents $-1.4 < I_{ext} < -0.3$, a stable limit cycle has appeared due to a supercritical Hopf bifurcation. The maximum and minimum voltages of this limit cycle are plotted as the stable points in this region of figure 7. The voltages corresponding to the unstable points running through the middle are acquired by finding the intersection of the nullclines, which in this regime gives the unstable fixed point. These points were deemed unstable by evaluating the eigenvalues of the stability matrix S at these fixed points. If at least one of the eigenvalues was positive, this point was labelled unstable. This regime where a stable limit cycle surrounds an unstable equilibrium is illustrated in the phase trajectories of figures 4 and 5. For currents $I_{ext} > -0.3$, another supercritical Hopf bifurcation results in the birth of a high-voltage stable state, and the simultaneous death of the stable limit cycle corresponding to tonic spiking of the neuron. This is a high-voltage steady state, demonstrated in the phase plane with figures 2 and 3.

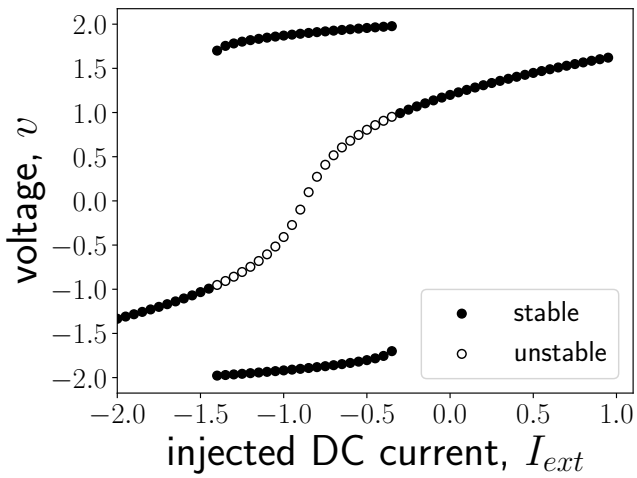

Figure 7: The equilibria of the FHN neuron are shown for each value of injected current. As this bifurcation diagram is read left-to-right, one may imagine slowly ramping up the injected current to a neuron that begins in its resting state, as shown in figure 8. Hopf bifurcations occur around $I_{ext} = -1.4$ and around $I_{ext} = -0.3$.

The same range of injected current that is reported in the bifurcation diagram of figure 7 is slowly ramped over in the time series of figure 8. This result shows how excitation block might actually look in an experiment where one had control of the injected current to some excitable cell and was able to simultaneously measure the membrane voltage. The model neuron here starts in the low-voltage state, slowly climbs the black dots on the left of figure 7, and then suddenly, at a critical current of $I_{ext} = -1.4$, starts to spike. After the injected current surpasses another critical value $I_{ext} = -0.3$, the neuron becomes boring again and tends to fixed values that are given by slowly following the black dots on the right of figure 7.

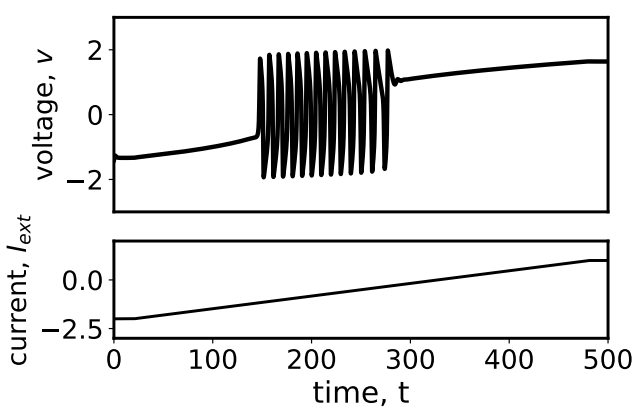

Figure 8: Demonstration of excitation block in response to a slow ramp of current spanning values shown in the bifurcation diagram of figure 7 above.

Noisy neuron

In this subsection, noise is added to the FitzHugh-Nagumo model in the following way:

$$\dot{v} = c\left(v - \frac{v^3}{3} + w + I_{\text{ext}}\right) + \zeta(t) \tag{5a}$$

$$\tau\dot{w} = -\frac{1}{c}(v - a + bw) + \eta(t) \tag{5b}$$

where ζ and η are Gaussian-distributed random variables with mean $\langle\zeta(t)\rangle = \langle\eta(t)\rangle = 0$ and auto-correlation $\langle\zeta(t)\zeta(s)\rangle = \langle\eta(t)\eta(s)\rangle = 2D\delta(t-s)$. The symbol δ represents the Dirac delta function, which may be defined as

$$\delta(t) \equiv \Theta'(t)$$

where Θ is the Heaviside step function

$$\Theta(t) \equiv \begin{cases} 0, & t < 0 \\ 1, & t \geq 1 \end{cases}$$

The reason that different symbols are used for the identically-distributed random variables ζ and η is because they are independent realizations of the Gaussian distribution. The Box-Mueller algorithm [13] is used here to solve equation 5 numerically. A numerical solution to the deterministic equations 1 is compared to a numerical solution of the stochastic equations 5 in figure 9. The noise appears to have a larger impact on the period of the spiking than the amplitude. This is likely due to the fact that noise is not only added to the voltage equation, but also to the evolution equation of the recovery variable w. The effect of noise on the evolution of w is shown in figure 10.

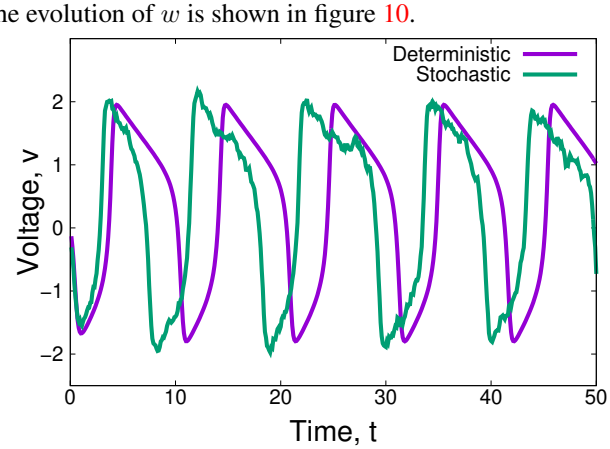

Figure 9: Deterministic versus stochastic FHN neuron for identical initial condition $v(0) = w(0) = 0.1$ and parameters $a = 0.7$, $b = 0.8$, $c = 3$, $I_{\text{ext}} = -0.5$. The amplitude of the noise in stochastic simulation is $D = 0.05$.

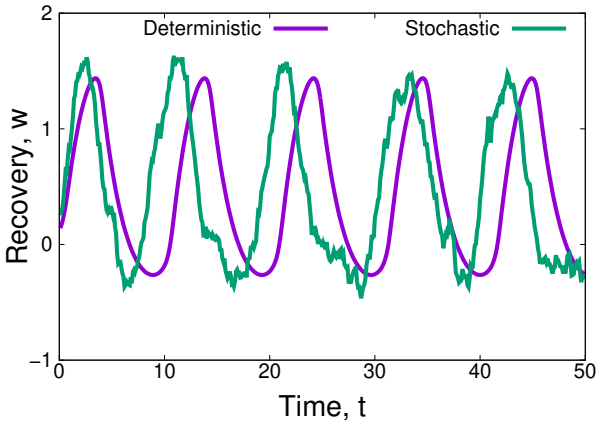

Figure 10: Deterministic versus stochastic FHN neuron with the same parameters as in figure 9, but here plotting the recovery variable w.

Threshold

The notion of a threshold of current whereby qualitatively different neuronal responses occur on either side of a critical I_{ext} is shown in figure 11. For low, negative currents the neuron is fated to an eternal low-voltage state, exemplified by the orange trace in figure 11. For higher external currents, a tonic spiking state is possible, as demonstrated with the blue trace where the neuron's voltage repetitively oscillates with constant frequency. As current is raised even higher, the large spiking oscillations turn into smaller damped oscillations about a new stable equilibrium. This is the high-voltage steady state that exists for high currents. The critical currents whereby these transitions occur are found approximately to be $I_{ext} = -1.4$ and $I_{ext} = -0.3$ from figure 7.

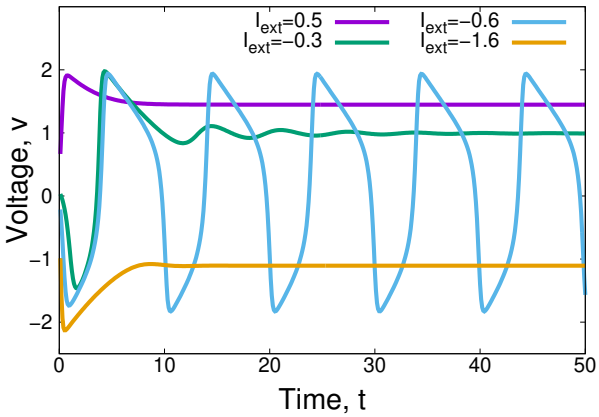

Figure 11: The FHN neuron response for various constant input currents I_{ext}, demonstrating the existence of a threshold. Each trace has the parameter values $v(0) = w(0) = 0.1$, $a = 0.7$, $b = 0.8$, $c = 3$.

Pair of electrically-coupled FHN neurons

Equation 4, describing a pair of electrically-coupled FHN neurons is studied numerically here. Figure 12 shows two neurons with asymmetric coupling $\gamma_{12} \neq \gamma_{21}$, initialized at different voltages. Both the voltage and the recovery traces have slight deformities with respect to their single-neuron counterparts. There are slight dips in the maxima of one neuron's voltage simultaneous with when the other neuron's voltage is minimal. Additionally, neuron 1 has a longer period of spiking than does neuron 2. This difference in period is also evident in the recovery variables, as well as an elevated maximum for neuron 2 over neuron 1. These asymmetries in the dynamics are due to the asymmetric coupling.

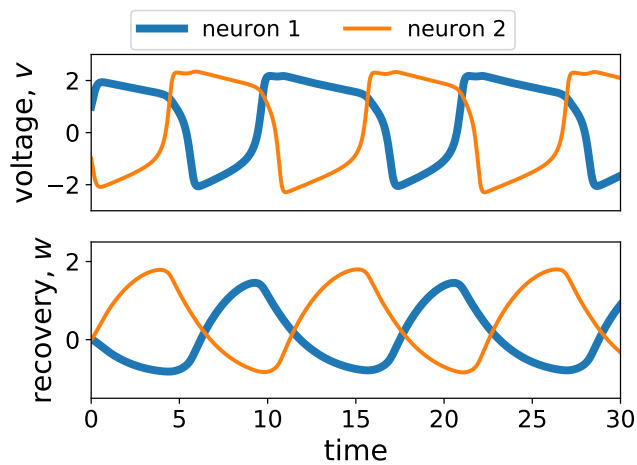

Figure 12: Two coupled FHN neurons with $\gamma_{12} = 0.5$, $\gamma_{21} = 1$ and initial conditions $v_1(0) = 1$, $v_2(0) = -1$, $w_1(0) = w_2(0) = 0$. The parameters a,b,c, and τ are the same as for previous sections.

If the asymmetry in the coupling is scaled up even further, it is possible for one neuron to greatly suppress the other neuron's spiking. This occurs suddenly as some cut-off coupling is reached, as depicted in the sequence of figures 13 and 14. In the traces of figure 13, the coupling γ_{21} is just strong enough for neuron 2 to follow its characteristic spiking trajectory, albeit with a reduced amplitude compared to neuron 1's voltage spikes. In figure 14 however, the coupling γ_{21} is so small with respect to γ_{12} that the difference in voltage puts neuron 2 on the right-hand side of the bifurcation diagram of figure 7. In other words, the effective "I_{ext}" for neuron 2 is the coupling term here, which is so positive in the case of figure 14 that it remains in the stable state given by the right-hand side of the bifurcation diagram. The small oscillations about this stable high-voltage state are on the same time-scale as neuron 1's pronounced spiking. This makes sense, since the input to neuron 2 is neuron 1's voltage.

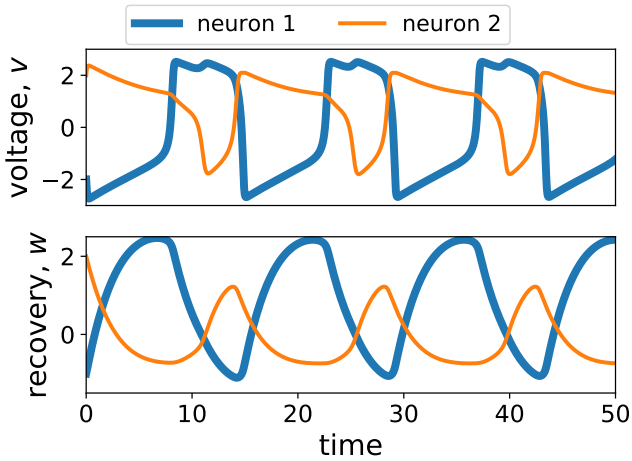

Figure 13: Two coupled FHN neurons with asymmetric coupling $\gamma_{12} = 2$, $\gamma_{21} = 0.2$.

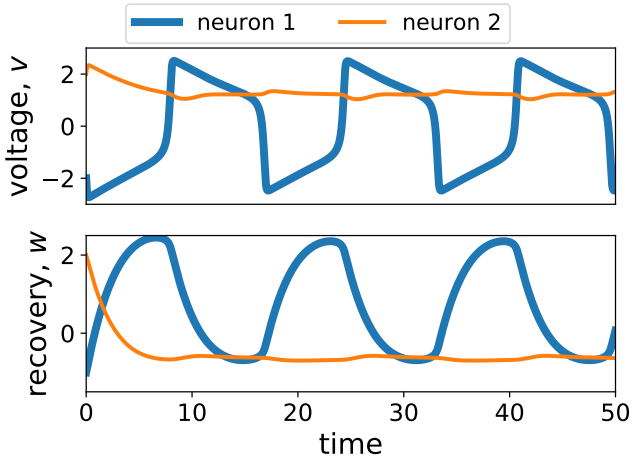

Figure 14: Two coupled FHN neurons with asymmetric coupling $\gamma_{12} = 2$, $\gamma_{21} = 0.1$.

Discussion

The famous model of FitzHugh and Nagumo for neural excitation has been contextualized historically, studied in its baseline form, related to the physiological phenomenon of excitation block, and extended to the scenarios of noisy inputs and two electrically-coupled neurons. It is inspiring and educational to look back on the history of this model, which developed on FitzHugh's side from a desire to understand the biophysically-realistic model of Hodgkin and Huxley at a deeper mathematical level. This is exemplified by the bifurcation diagram shown in figure 7, which illustrates a very characteristic behavior of neurons whereby a spiking state emerges and disappears discontinuously with respect to some parameter (here, I_{ext}). Although not explored here, neurons that undergo Hopf bifurcations with respect to the injected DC current have a specific scaling relationship between their frequency and stimulus strength, which Hodgkin called "class 2 excitability" [14]. This is distinct from "class I excitability," whereby the neuron instead displays an infinite-period bifurcation. FitzHugh was a pioneer in this mathematical approach to neuroscience. Nagumo made the extension of equation 1 to spatially-extended excitable systems, which the brain and heart comprise. He tackled this problem with an engineering approach, producing a circuit with nonlinear elements called tunnel diodes (invented by the Japanese physicist Leo Esaki) that displays analogous dynamics to the mathematical model developed by FitzHugh. Nagumo considered the propagation of action potentials along the length of the axon, a protruding part of a neuron that sends messages to other neurons. He did so by formulating the Hodgkin-Huxley equations as a system of partial differential equations, showing that it has traveling wave solutions.

Acknowledgements

Much of the historical context described here has been adapted from the open-access online encyclopedia Scholarpedia.

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